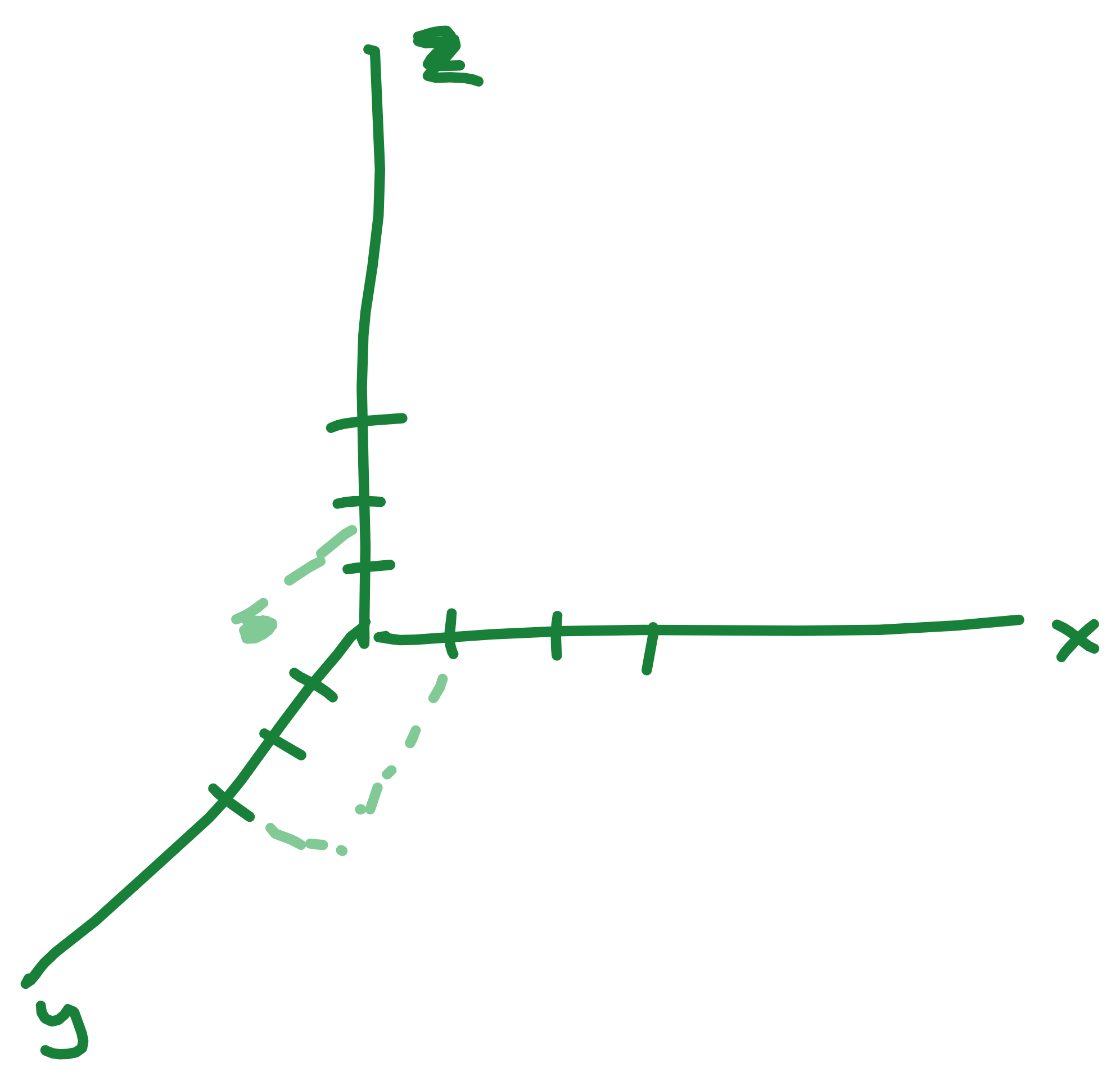
Space Coordinates

Section 11.2

8-14-26

Ex) Draw $\langle 1, 3, 2 \rangle$



A) Extensions

1. Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$

2. Distance: $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

3. Sphere: $(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$

B) Determine Co-Linear Points

Ex) P, Q, R

$\langle P, Q \rangle$ has to be a multi of $\langle P, R \rangle$

OR

$\langle P, R \rangle$ has to be a multi of $\langle P, Q \rangle$

29) A: $\langle -1, 0, -2 \rangle$

B: $\langle -1, 5, 2 \rangle$

C: $\langle -3, -1, 1 \rangle$

Dist. $A \leftrightarrow B = \sqrt{(0)^2+(5)^2+(4)^2}$

$= \sqrt{41}$

Dist $B \leftrightarrow C = \sqrt{(2)^2+(-6)^2+(-1)^2} = \sqrt{41}$

Dist $C \leftrightarrow A = \sqrt{(2)^2+(1)^2+(-3)^2} = \sqrt{14}$

$\sqrt{41}, \sqrt{41}, \sqrt{14}$

Isoceles

53) $\langle 3, 1, 8 \rangle$

61) a) $\langle -6, -4, 10 \rangle$
b) $\langle 2, 4/3, -30/3 \rangle$ Both Parallel

67) A: $\langle 1, 2, 4 \rangle$ B: $\langle 2, 5, 0 \rangle$ C: $\langle 0, 1, 5 \rangle$

$\langle A, B \rangle = \langle 1, 3, -4 \rangle$

$\langle A, C \rangle = \langle -1, -1, 1 \rangle$

$\boxed{\text{Not Collinear}}$

69) AB $\langle 1, 2, 3 \rangle$

BC $\langle -3, -1, -2 \rangle$

CD $\langle 1, 2, 3 \rangle$

DA $\langle 1, -3, -4 \rangle$

Parallel

$\boxed{\text{Not parallelogram}}$

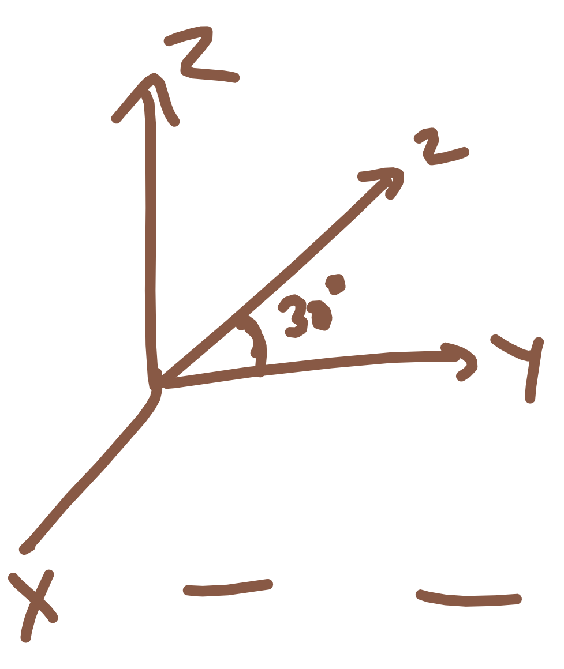
79) $\langle 4, -5, 3 \rangle$ $\sqrt{16+25+9} = \sqrt{50}$

a) $\frac{1}{\sqrt{50}} v$ b) $-\frac{1}{\sqrt{50}} v$

85) $\sqrt{4+4+1} = \sqrt{9} = 3$

$\frac{1}{3} \cdot \frac{3}{2} (u)$ or $\frac{1}{2} u$

87)



$\langle 0, 2\cos(30)\, 2\sin(30) \rangle$

$\langle 0, \sqrt{3}, 1 \rangle$